TAMBAN MODEL 2

Three-Architecture Retraining and Internal Development Evaluation

YOLO26m | Faster R-CNN ResNet50-FPN V2 | EfficientNetV2-M + RetinaNet

Development-stage result

YOLO26m produced the highest internal-development mAP50-95 (0.4606) and the lowest measured detector latency (54.6 ms/image). Faster R-CNN V2 remained very competitive, with a marginally higher F1 score at its own tuned threshold. The EfficientNetV2-M RetinaNet configuration produced zero evaluated detection performance and must be treated as a failed experimental configuration rather than evidence that the EfficientNet family is inherently unsuitable.

846	11,620	12	3
Cleaned images	Part boxes	Grade-part classes	Architectures

Project	Dried Fish Models - Tamban quality inspection
Scope	Model 2 only: grade-aware Head / Body / Tail detection
Dataset	846 cleaned images; 11,620 annotated part boxes; 12 detector classes
Evaluation split	680 train / 166 internal development images; group-aware engineering split
Document status	Development evaluation report - not final specimen-independent thesis test

Prepared: 25 September 2026

1. Executive Summary

This experiment retrained and compared three separate Model 2 object detectors on the same cleaned Tamban part-annotation pool. The objective was not whole-fish counting; it was grade-aware detection of anatomical regions so that the downstream system can collect evidence for Class A, Class B, Class C, and Rejected fish. Each model therefore attempted the same 12-class detection task: four quality families multiplied by Head, Body, and Tail.

> **Primary engineering interpretation**
>
> For the next integration stage, YOLO26m is the strongest first candidate because it combined the highest mAP50-95 (0.4606), highest mAP50 (0.7876), slightly higher recall (0.8625), and substantially lower detector latency (54.6 ms/image). Faster R-CNN V2 should be retained as a serious comparator because its precision and F1 were slightly higher at its own tuned confidence threshold. This is a development-stage engineering choice, not a final thesis conclusion.

Key results:

- YOLO26m: mAP50-95 0.4606; mAP50 0.7876; precision 0.8018; recall 0.8625; F1 0.8310; latency 54.6 ms/image.
- Faster R-CNN V2: mAP50-95 0.4405; mAP50 0.7665; precision 0.8122; recall 0.8553; F1 0.8331; latency 123.7 ms/image.
- EfficientNetV2-M RetinaNet: all evaluated detection metrics were 0.0000 despite a completed training run; the configuration requires debugging before any architectural conclusion is drawn.
- The weakest recurring anatomical region was the tail, especially Grade_B_Tail. Bodies and heads were generally much stronger.
- These are detector results on an internal development split. They are not final fish-level grading accuracy and are not a specimen-independent test result.

Document Map

- Experiment objective and Model 2 role
- Dataset, labels, split strategy, and training setup
- Evaluation metrics and interpretation rules
- Overall quantitative comparison
- Detailed YOLO26m, Faster R-CNN, and EfficientNet analyses
- Per-class behavior, latency, and training-loss analysis
- Implications for live grading and the Ungraded state
- Limitations, next validation plan, and thesis-ready wording
- Checkpoint and artifact registry

2. Experiment Objective and Model 2 Role

Model 2 is the quality-evidence detector in the two-model Tamban inspection system. It does not count whole fish. Instead, it detects grade-aware anatomical parts so that the later software stages can associate those detections with individual fish and combine the evidence into a final per-fish decision.

Input image	**Model 2 detector**	**12 part-grade boxes**	Fish-part association	Final fish grading

Context flow: this report evaluates only the Model 2 detector stage; association and final per-fish grading are downstream stages.

2.1 Detector Classes

Grade family	Body class	Head class	Tail class
Class A	Grade_A_Body	Grade_A_Head	Grade_A_Tail
Class B	Grade_B_Body	Grade_B_Head	Grade_B_Tail
Class C	Grade_C_Body	Grade_C_Head	Grade_C_Tail
Rejected	Rejected_Body	Rejected_Head	Rejected_Tail

The detector therefore learns both anatomy and quality-family labels simultaneously. A successful prediction such as Grade_C_Body means that the model has localized a body region and assigned that region to the Class C quality family. A final fish-level verdict still requires downstream association and fusion; the detector output alone is not the final grade.

> **Important distinction**
>
> Ungraded is not one of the 12 Model 2 detector classes. It is a downstream abstention state used when the complete system does not have enough trustworthy, correctly associated evidence to issue a fish-level grade.

3. Dataset and Experimental Design

The completed r03 experiment used the cleaned Model 2 training pool and forced all three architectures to solve the same detection task against the same internal development split. This is essential for a meaningful architecture comparison: the reported metric differences are not caused by evaluating each model on a different development set.

Images	846 cleaned images
Annotations	11,620 part bounding boxes
Classes	12 grade-part classes
Training subset	680 images
Internal development subset	166 images
Development fraction	Approximately 20%
Split purpose	Engineering model development and comparison; not a final specimen-independent test

3.1 Group-aware internal development split

The single-cell training pipeline did not use a simple random image split. It first removed Roboflow-style hash suffixes from filenames, then grouped time-coded WIN captures into two-minute bins so that nearby captures were more likely to stay together. For non-WIN images, the source stem was used as the grouping key. The code tried many GroupShuffleSplit candidates and selected a split that kept groups separated while preserving representation of all 12 classes as closely as possible.

This is an improvement over an ordinary random split because near-duplicate or temporally adjacent captures are less likely to be scattered between training and development sets. However, the split is still explicitly labeled internal development because physical specimen identity is not available for every image. It therefore cannot prove performance on completely unseen physical fish.

3.2 Training configuration

Architecture	Epochs	Input size	Pretraining	Optimizer / LR	Sampling	Key augmentation
YOLO26m	70	768	Pretrained YOLO26m	AdamW, lr0=0.001	Native YOLO loader	Mosaic 0.5; close mosaic 10; mild HSV; rotation/translation /scale; H/V flips
Faster R-CNN ResNet50-FPN V2	40	min 768 / max 1024	Torchvision DEFAULT detection weights	AdamW, lr=0.0002	Rare-class-aware weighted image sampling	H/V flips; 90-degree rotation; mild affine transforms
EfficientNetV2-M + FPN + RetinaNet	40	min 768 / max 1024	Pretrained EfficientNetV2-M backbone	AdamW, lr=0.00015	Rare-class-aware weighted image sampling	Same mild spatial augmentation; BatchNorm frozen during training

For the torchvision detectors, the weighted sampler used an inverse-square-root class-frequency strategy capped at a maximum image weight of 3.0. This intentionally gives rare classes more exposure without applying extreme inverse-frequency oversampling. YOLO used modest color augmentation because brightness, saturation, and reflectance can be meaningful quality cues in the fish images; the training recipe therefore avoided strong hue manipulation.

3.3 EfficientNet detector construction

EfficientNetV2-M is normally a classification backbone rather than a complete object detector. In this experiment it was converted into a detector by extracting the final four spatial-resolution stages, feeding them into a Feature Pyramid Network (FPN), and attaching a RetinaNet head. The custom RetinaNet used five anchor scales (24, 48, 96, 192, 384 pixels) and multiple aspect ratios (0.4, 0.6, 1.0, 1.7, 2.5). Because this is a custom construction, a zero-result experiment must be debugged as an implementation/training configuration before the underlying EfficientNet family is judged.

4. Evaluation Metrics and How to Read Them

Metric	Meaning	Interpretation in this experiment
mAP50-95	Mean average precision averaged across IoU thresholds 0.50 to 0.95.	Primary localization-quality summary; stricter than mAP50 and sensitive to box precision.
mAP50	Mean average precision at IoU 0.50.	Shows whether the model finds and classifies parts with a more forgiving localization requirement.
mAP75	Mean average precision at IoU 0.75.	Tests tighter box localization.
Precision@bestF1	Fraction of accepted predictions that are correct at the development-tuned F1 threshold.	Higher precision means fewer false-positive part detections at that threshold.
Recall@bestF1	Fraction of ground-truth objects detected at the development-tuned F1 threshold.	Higher recall means fewer missed anatomical parts.
F1@IoU0.5	Harmonic mean of precision and recall at IoU 0.50.	Useful balance metric, but only at the tuned development threshold.
Dev confidence threshold	Score threshold selected on the development set to maximize F1.	Architecture-specific operating point; raw values should not be compared as if confidence were calibrated across models.
Inference latency	Average detector time per image on the current Colab environment.	Hardware- and implementation-dependent; useful for relative deployment comparison.

Metric	Meaning	Interpretation in this experiment

Do not call mAP50 "accuracy"

mAP50, mAP50-95, precision, recall, and F1 describe different aspects of object detection. They are not interchangeable with final fish-grade classification accuracy. A fish-level accuracy value can only be calculated after parent-fish association, evidence fusion, and a ground-truth final grade are available.

5. Overall Quantitative Results

Model	mAP50-95	mAP50	mAP75	Precision	Recall	F1	Dev conf	Latency ms	Train min
YOLO26m	0.4606	0.7876	0.4636	0.8018	0.8625	0.8310	0.35	54.6	33.3
Faster R-CNN V2	0.4405	0.7665	0.4564	0.8122	0.8553	0.8331	0.70	123.7	63.6
EfficientNet V2-M RetinaNet	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.05	140.2	41.8

YOLO26m exceeded Faster R-CNN V2 by 2.02 percentage points in mAP50-95 and 2.12 percentage points in mAP50. Faster R-CNN V2 was higher by 1.04 percentage points in precision and 0.21 percentage points in F1. These differences show that the two successful detectors are close in overall detection quality rather than being separated by a large accuracy gap.

Latency creates a much larger practical separation. YOLO26m measured 54.6 ms/image versus 123.7 ms/image for Faster R-CNN V2. Under the measured detector-only timing, Faster R-CNN was about 2.26x slower, while YOLO reduced latency by approximately 55.8% relative to Faster R-CNN. The corresponding detector-only throughputs are approximately 18.3 and 8.1 images/second, before Model 1, association, UI rendering, or any other application overhead.

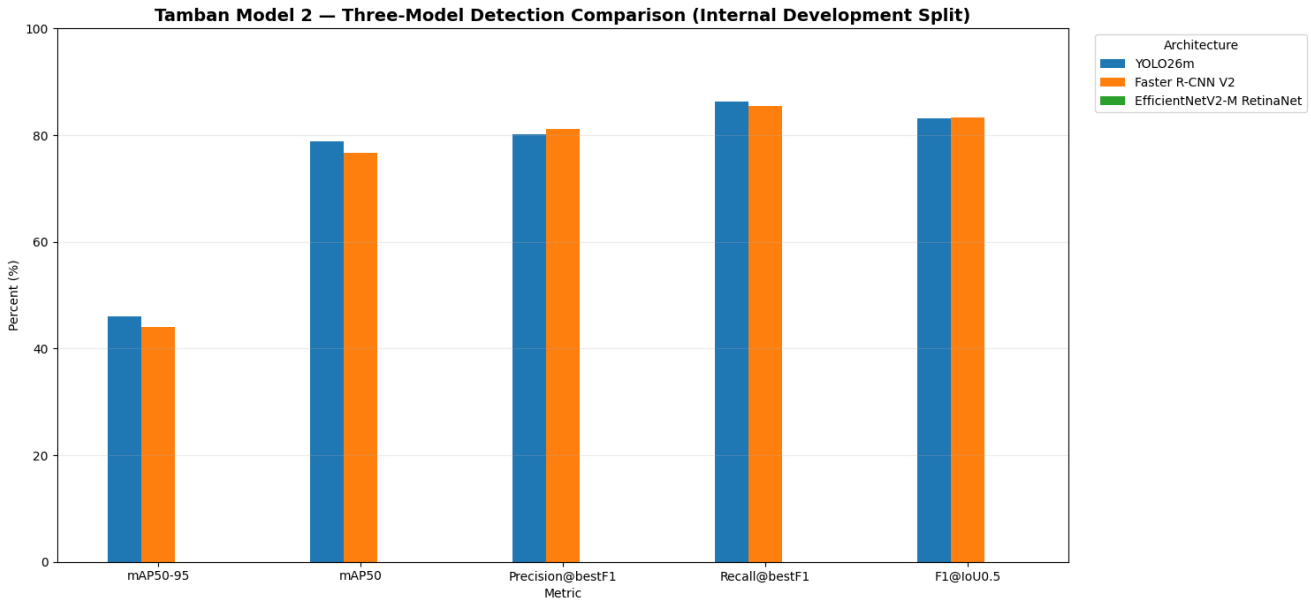


Figure 1. Overall internal-development comparison of YOLO26m, Faster R-CNN V2, and EfficientNetV2-M RetinaNet. EfficientNet values are zero because the evaluated configuration produced no valid detection performance.

6. YOLO26m Detailed Analysis

YOLO26m produced the strongest combined internal-development profile. Its mAP50 of 0.7876 indicates that many grade-part objects were detected and classified correctly at IoU 0.50, while its mAP50-95 of 0.4606 shows a substantial drop as localization requirements become stricter. This gap is expected to matter most for small or narrow anatomical targets because minor box displacement causes a larger IoU penalty.

The tuned development confidence threshold was 0.35. This threshold should be treated as a candidate operating point for direct detector testing, not as a universal permanent threshold. It was selected to maximize F1 on the internal development set and therefore must be rechecked on newly captured validation images.

Class	AP50	AP50-95	Observation
Grade_A_Body	99.4%	65.5%	Very strong AP50
Grade_A_Head	96.9%	71.8%	Very strong AP50
Grade_A_Tail	84.8%	45.2%	Strong AP50
Grade_B_Body	73.7%	33.6%	Moderate-good AP50
Grade_B_Head	86.7%	37.7%	Strong AP50
Grade_B_Tail	35.9%	16.1%	Weak AP50; priority for improvement
Grade_C_Body	97.0%	65.1%	Very strong AP50
Grade_C_Head	96.2%	69.2%	Very strong AP50
Grade_C_Tail	64.6%	27.0%	Limited AP50
Rejected_Body	86.5%	49.0%	Strong AP50
Rejected_Head	82.0%	53.6%	Strong AP50
Rejected_Tail	50.3%	25.0%	Limited AP50

The exact YOLO26m class breakdown shows why the global score alone is insufficient. Grade_A_Body, Grade_A_Head, Grade_C_Body, and Grade_C_Head all exceeded 96% AP50, while Grade_B_Tail was only 35.9% AP50. Grade_C_Tail and Rejected_Tail were also substantially weaker than their corresponding body/head detections. This makes tail evidence the clearest anatomical bottleneck in the current Model 2 detector.

7. Faster R-CNN V2 Detailed Analysis

Faster R-CNN V2 also learned the 12-class detection task successfully. Its mAP50-95 was 0.4405 and mAP50 was 0.7665, placing it close to YOLO26m. At its own F1-optimized development threshold, Faster R-CNN had the highest precision (0.8122) and the highest F1 (0.8331), but these improvements were small: F1 was only about 0.21 percentage points above YOLO26m.

The architecture required a much higher tuned score threshold (0.70) than YOLO26m (0.35). This does not mean Faster R-CNN is objectively "more confident." Detector scores from different architectures are not calibrated to the same scale. The threshold simply identifies the operating point that maximized F1 for that specific model on this specific development set.

From the per-class AP50 chart, Faster R-CNN appears slightly stronger on some head/body categories, including Grade_B_Head, Grade_C_Body, Grade_C_Head, Rejected_Head, and Rejected_Tail. YOLO26m appears stronger on several tail and difficult B/C-related regions, especially Grade_A_Tail, Grade_B_Body, Grade_B_Tail, and

Grade_C_Tail. These are chart-level comparative observations; exact Faster R-CNN per-class values should be read from the generated model2_per_class_ap50.csv before they are quoted numerically in the thesis.

8. EfficientNetV2-M RetinaNet Failure Analysis

> **Experimental failure state**
>
> The EfficientNetV2-M RetinaNet configuration completed a training run but returned mAP50-95 = 0, mAP50 = 0, precision = 0, recall = 0, and F1 = 0. This is not a normal "weaker model" result. It indicates that the evaluated configuration failed to produce useful detections under the implemented pipeline.

Because the model is a custom detector assembled from an EfficientNetV2-M backbone, a Feature Pyramid Network, and RetinaNet, multiple failure points are possible. The current experiment does not identify the root cause. It would be scientifically incorrect to conclude that EfficientNet itself is incapable of this task based solely on this one failed configuration.

Recommended debugging sequence before any retraining:

1. Run raw inference on 5-10 known training images at an extremely low score threshold (for example 0.001) and inspect whether any boxes are produced.
2. Intentionally overfit a tiny 5-10 image subset. If the detector cannot memorize that subset, suspect implementation, target formatting, anchors, feature mapping, or optimization rather than dataset generalization.
3. Verify foreground/background label indexing for RetinaNet and confirm that the 12 foreground classes are mapped consistently during training and evaluation.
4. Inspect classification and box-regression losses independently rather than only total normalized loss.
5. Verify FPN feature-map resolutions and channel dimensions, anchor scales/aspect ratios, and target-box scale relative to the 768-1024 input range.
6. Confirm input normalization and resizing are consistent between training and inference.
7. Only after the tiny-subset test succeeds should a full EfficientNetV2-M RetinaNet retraining be considered.

9. Per-Class AP50 Comparison

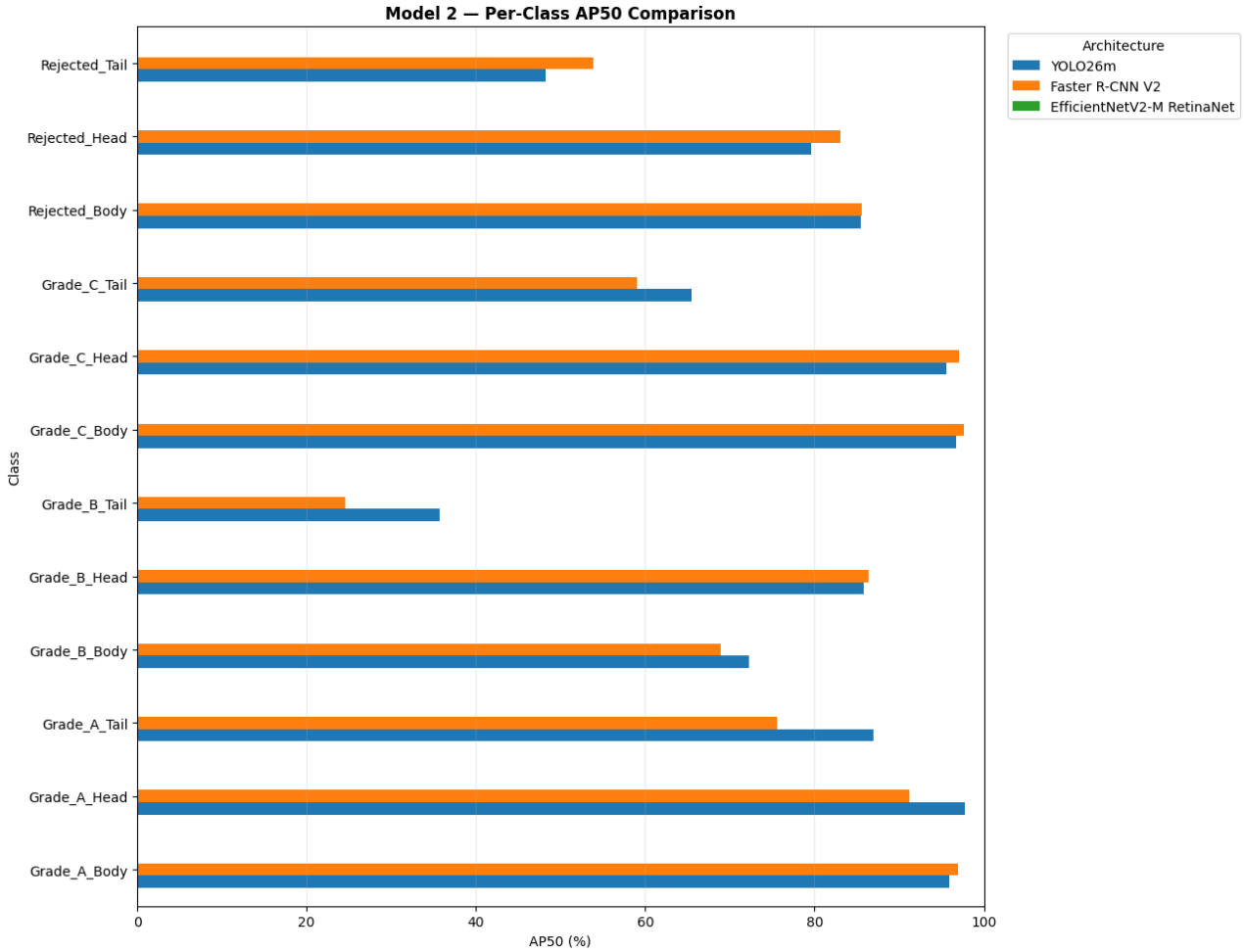


Figure 2. AP50 by grade-part class. The chart exposes the anatomical and quality families that are hidden by one overall mAP value.

9.1 Grade-family interpretation

Grade family	Body behavior	Head behavior	Tail behavior
Class A	Both strong; Faster R-CNN slightly higher in chart	YOLO26m visibly higher	YOLO26m visibly higher
Class B	YOLO26m higher	Very close; Faster R-CNN slightly higher	Weakest region for both; YOLO26m clearly higher
Class C	Both near top; Faster R-CNN slightly higher	Both near top; Faster R-CNN slightly higher	YOLO26m higher, but still much weaker than C body/head
Rejected	Almost identical	Faster R-CNN higher	Both limited; Faster R-CNN higher in chart

The most important pattern is not which model is ahead by a few points on one easy class. It is that tails are systematically harder than bodies and heads. This matters directly to the final grading system: if the software requires a reliable tail on every fish before it can grade, the weakest region will control the entire system and inflate the Ungraded rate.

10. Accuracy-Latency Trade-off

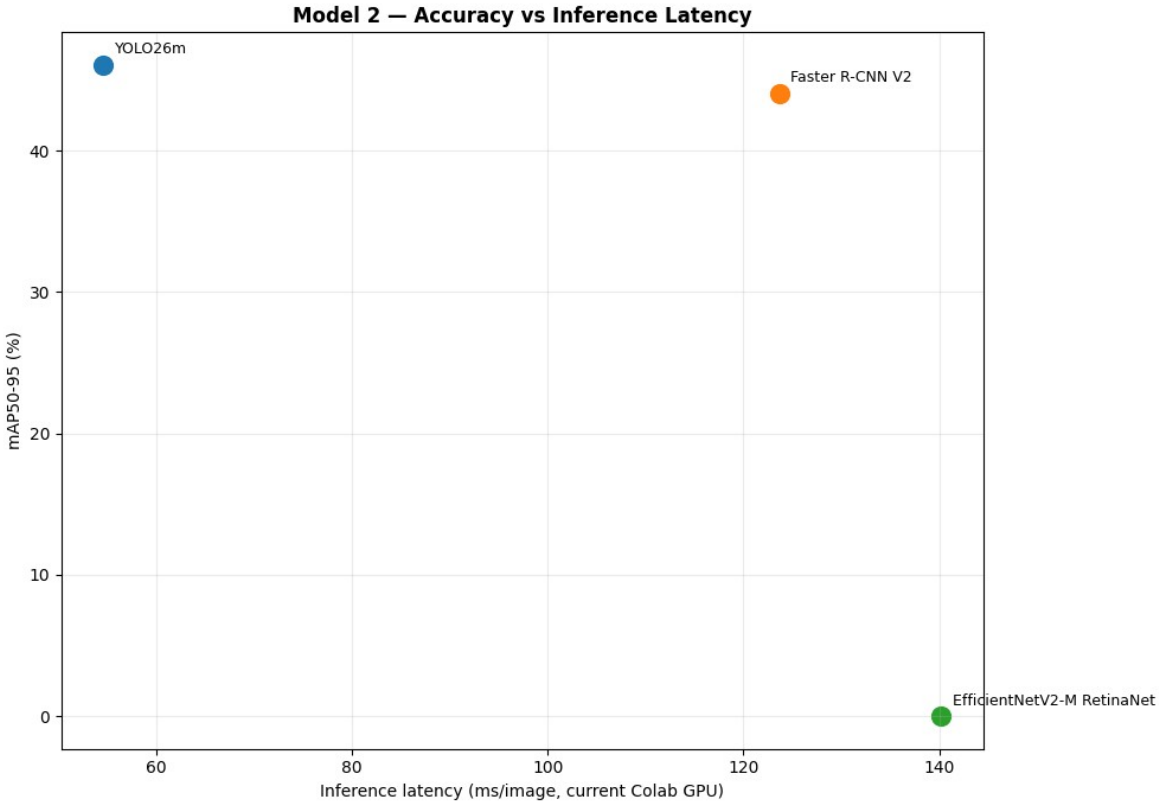


Figure 3. mAP50-95 versus inference latency. Upper-left is preferable for a live detector: higher localization performance with lower latency.

Model	Latency (ms/image)	Approx. detector-only throughput	Deployment interpretation
YOLO26m	54.6	18.3 img/s	Fastest successful detector; best first candidate for live integration testing
Faster R-CNN V2	123.7	8.1 img/s	Strong quality comparator but substantially slower
EfficientNetV2-M RetinaNet	140.2	7.1 img/s	Slowest and no usable detection performance in current configuration

The throughput values above are simple reciprocals of the measured per-image detector latency. They are not end-to-end application FPS. The complete system will also spend time on Model 1, part-to-fish association, feature extraction, tracking, fusion, UI drawing, and camera I/O. The latency results should therefore be interpreted as relative detector cost rather than a guaranteed live-system frame rate.

11. Training Loss and Convergence Behavior

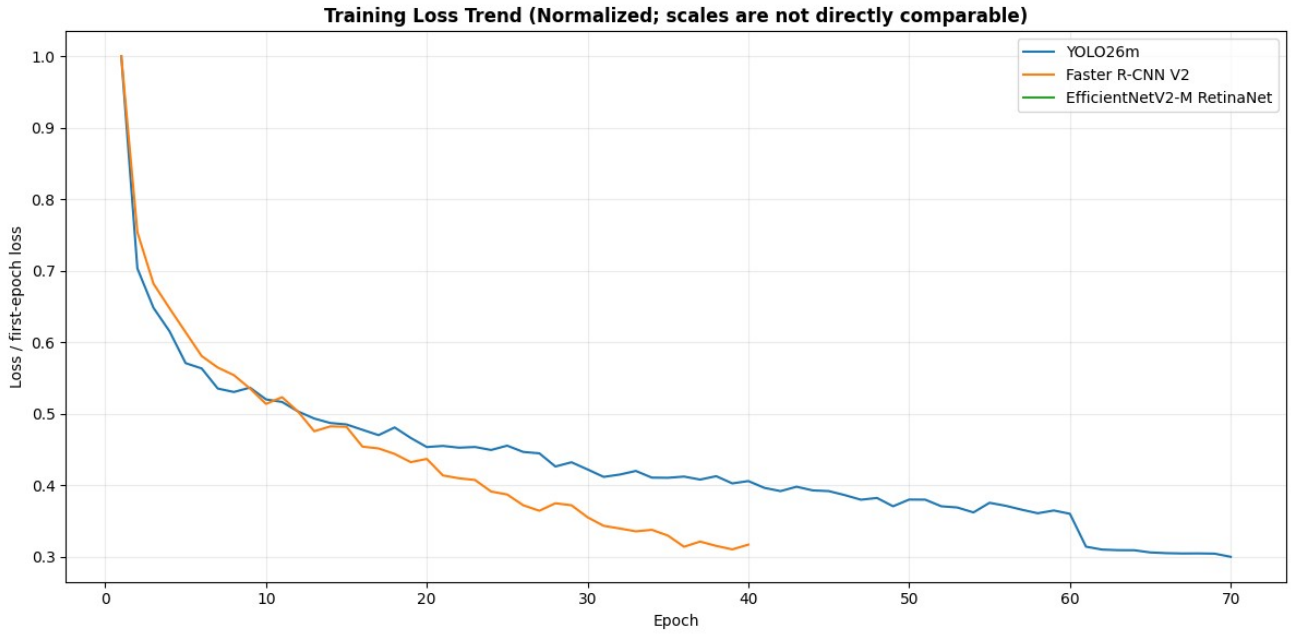


Figure 4. Normalized training-loss trend. Each curve is divided by its first-epoch loss, so only within-model trend should be interpreted; raw loss magnitudes are not comparable across architectures.

YOLO26m and Faster R-CNN V2 both show a sustained downward normalized-loss trend rather than numerical divergence. YOLO decreases rapidly early, then more gradually, and shows another clear drop after approximately epoch 60. In the training log, mosaic augmentation was closed near this point, so the late-stage loss change is consistent with the planned training-regime transition. Faster R-CNN decreases more steadily across its 40 epochs.

The EfficientNet curve is not visibly present in the exported normalized-loss plot even though the legend includes the architecture. Combined with its all-zero evaluation metrics, this figure provides no evidence of a healthy useful detector trajectory for that configuration. Training loss alone, however, cannot identify the exact failure mechanism.

> **Loss caution**
>
> A falling training loss does not equal final accuracy and does not prove generalization. Overfitting, leakage, and capture-session memorization can still occur even when the training curve looks smooth. Independent validation remains required.

12. Confidence Thresholds and Downstream Grading Implications

The development sweep selected 0.35 for YOLO26m and 0.70 for Faster R-CNN V2. These thresholds should be carried into controlled direct-inference tests as starting points, but they should not be treated as permanent production constants. A new specimen-independent validation set may select different values.

For this project, recall has unusual importance because a missed part does more than reduce detector recall: it also removes evidence from the later fish-level grading stage. At the same time, lowering the threshold too aggressively can create false part boxes and incorrect fish-part associations. The final operating point therefore needs to balance detector precision/recall together with downstream grading coverage and false-grade risk.

12.1 Why tail weakness matters to the 50/30/20 evidence design

The current grading concept weights Body at 50%, Head at 30%, and Tail at 20%. Because tail detection is the least reliable anatomical region, a strict rule that demands Head + Body + Tail for every fish would allow the weakest

component to determine whether the entire fish can be graded. A more defensible design is to keep missing evidence unrenormalized and validate an evidence-coverage rule on independent data.

Example	Body contribution	Head contribution	Tail contribution	Total support / coverage
Strong B body + head; tail unavailable	0.93 x 50 = 46.5	0.88 x 30 = 26.4	0	72.9 / 80%
Body only	0.82 x 50 = 41.0	0	0	41.0 / 50%

The table is an illustration of evidence accounting, not a validated decision threshold. The final minimum coverage, score, and winner-margin rules must be tuned on independent validation fish. Missing or occluded evidence should not automatically be converted into Rejected evidence.

13. Scientific Limitations of the Current Comparison

The r03 experiment is useful for architecture development, but its scientific scope must be stated precisely. The strongest limitations are listed below.

- Internal development rather than final test: the 166-image split was used for model development and threshold tuning.
- No complete physical specimen identity registry: group-aware splitting reduces nearby-capture leakage risk but cannot guarantee that every physical fish is isolated to one split.
- Detector metrics are not fish-level grade accuracy: the experiment evaluates 12 part classes, not the final A/B/C/Rejected/Ungraded decision per parent fish.
- Per-class imbalance remains relevant: difficult/less represented anatomical regions can be hidden by strong performance on common body/head classes.
- Capture-context effects may remain: background, lighting, reflectance, orientation, and repeated capture conditions can contribute to development-set performance.
- Latency is hardware-specific: the measured ms/image values apply to the current Colab runtime and are not guaranteed on the final deployment machine.
- EfficientNet result is configuration-specific: a zero-result custom RetinaNet run is not sufficient evidence to reject the EfficientNet family in general.
- No final association evaluation is included: the experiment does not measure whether Head/Body/Tail detections are attached to the correct fish in crowded scenes.

> **Correct reporting language**
>
> Use phrases such as "internal-development detection performance" or "development-set mAP." Do not describe the 0.7876 YOLO mAP50 as "78.76% final system accuracy."

14. Recommended Next Validation Plan

The next stage should stop changing the model weights temporarily and test whether the new checkpoints behave correctly in direct inference and in the application pipeline. This isolates training quality from integration errors.

1. Freeze and archive the r03 YOLO26m and Faster R-CNN V2 dev-best checkpoints, plus the split CSV and model registry.
2. Direct-training-image sanity test: run Model 2 alone on exact known training images. A model that cannot detect its own learned patterns at low-to-moderate thresholds requires investigation before integration.
3. Internal-development visual audit: inspect representative A/B/C/Rejected images, including tails and overlap cases, and compare predicted classes/boxes with labels.

4. New-image generalization test: capture entirely new fish images that were not part of the cleaned pool. Include mixed grades in the same scene when possible.
5. Compare YOLO26m and Faster R-CNN on exactly the same new images; report per-class precision/recall/AP and especially B-versus-C behavior.
6. Integration A/B test: run Model 2 on the original full frame and compare against Model 1 crop -> Model 2 using the same images. Do not assume crop inference is superior without evidence.
7. Evaluate part-to-fish association with reviewed parent IDs in crowded scenes; allow ambiguous parts to remain unassigned instead of forcing nearest-parent ownership.
8. Tune final evidence-coverage, score, and ambiguity thresholds using a validation set only.
9. Lock the full pipeline, then evaluate a separate untouched final test set once for thesis reporting.

14.1 Minimum direct-inference test matrix

Test case	Purpose	Models	Expected diagnostic value
Exact training image	Confirm learned detector is not dead	YOLO + Faster R-CNN	Failure points to checkpoint/preprocessing/model-loading problem
Internal dev image	Check development behavior visually	YOLO + Faster R-CNN	Confirms metrics correspond to sensible boxes/classes
Completely new image	Check generalization	YOLO + Faster R-CNN	Exposes dataset/capture-session dependence
Overlapping fish	Stress part separation	YOLO + Faster R-CNN	Exposes ownership ambiguity and tail/head confusion
Full frame vs Model 1 crop	Check integration distribution shift	Chosen detector	Separates detector weakness from crop/resizing handoff issues

15. Suggested Thesis Wording for the Current Development Result

Development-results paragraph

On the 166-image internal development split, YOLO26m achieved an mAP50 of 0.7876 and an mAP50-95 of 0.4606, while Faster R-CNN ResNet50-FPN V2 achieved 0.7665 and 0.4405, respectively. Faster R-CNN obtained a marginally higher F1 score at its development-tuned confidence threshold (0.8331 versus 0.8310), whereas YOLO26m produced higher average precision, slightly higher recall, and substantially lower inference latency (54.6 ms/image versus 123.7 ms/image). The EfficientNetV2-M RetinaNet configuration produced zero evaluated detection performance under the implemented training configuration and was treated as an experimental failure requiring further diagnosis. These values represent internal-development object-detection performance for 12 grade-part classes and are not final fish-level classification accuracy.

15.1 Defense-ready interpretation

Question	Defensible answer
Why does YOLO26m move forward first?	Because it produced the highest mAP50-95 and mAP50 while also being substantially faster. Faster R-CNN remains a valid comparator because its precision and F1 were slightly higher at its own tuned threshold.

Question	Defensible answer
Why not call YOLO the final best model?	The comparison uses an internal development split. A final selection requires new specimen-independent validation and then an untouched test set.
Why is Faster R-CNN still important?	It achieved very similar detection quality and provides an architectureally different two-stage benchmark. If new-data behavior differs from YOLO, that comparison is scientifically informative.
Why did EfficientNet get zero?	The custom EfficientNetV2-M + FPN + RetinaNet detector failed under the evaluated configuration. The experiment does not identify whether the cause is optimization, anchors, feature mapping, label handling, preprocessing, or another implementation detail.
What is the main class-level weakness?	Tail detection, particularly Grade_B_Tail, is the clearest weakness. The result supports careful evidence-coverage logic rather than demanding a perfectly detected tail for every fish.
Does 78.76% mAP50 mean 78.76% final accuracy?	No. mAP50 is an object-detection metric. Final fish-grade accuracy must be measured after part association and final decision fusion against fish-level ground truth.

16. Checkpoint and Output Artifact Registry

The completed single-cell run generated the following reproducibility artifacts in the Model 2 r03 comparison folder.

Artifact	Filename	Role / handling
Primary YOLO checkpoint	model2_tamban_yolo26m_partdet_r03_devbest.pt	Retain/freeze for next-stage testing
YOLO last checkpoint	YOLO training run weights/last.pt	Training endpoint; dev-best remains preferred for comparison
Faster R-CNN dev-best	model2_tamban_fasterrcnn_r50fpn_v2_partdet_r03_devbest.pth	Retain/freeze as comparison candidate
Faster R-CNN last	model2_tamban_fasterrcnn_r50fpn_v2_partdet_r03_last.pth	Final epoch checkpoint
EfficientNet dev-best	model2_tamban_effnetv2m_retinanet_partdet_r03_devbest.pth	Experimental failed configuration; do not deploy
EfficientNet last	model2_tamban_effnetv2m_retinanet_partdet_r03_last.pth	Final epoch checkpoint of failed configuration
Comparison metrics	model2_three_model_comparison.csv	Overall architecture metrics
Per-class metrics	model2_per_class_ap50.csv	Exact AP50 per class for architecture comparison
Split manifest	model2_r03_internal_development_split.csv	Records train/dev membership and grouping
Model registry	model2_r03_model_registry.json	Names/checkpoint metadata
Generated plot 01	01_model2_three_model_quality_comparison.png	Overall metric comparison
Generated plot 02	02_model2_accuracy_vs_speed.png	Accuracy-latency comparison
Generated plot 03	03_model2_per_class_ap50_comparison.png	Per-class AP50 comparison

Artifact	Filename	Role / handling
Generated plot 04	04_model2_normalized_training_loss.png	Normalized loss trends
Readme	READ_ME_AFTER_TRAINING.txt	Post-training notes generated by the experiment

16.1 Naming convention interpretation

The naming convention encodes the system role, species/project context, architecture, task, revision, and checkpoint purpose. For example, model2_tamban_yolo26m_partdet_r03_devbest.pt means: Model 2; Tamban project; YOLO26 medium architecture; part-detection task; revision 03; best checkpoint selected by internal-development performance.

17. Conclusion

The r03 Model 2 retraining experiment successfully produced two usable, competitive part detectors and one failed custom detector configuration. YOLO26m achieved the highest internal-development mAP50-95 and mAP50 while also providing the lowest measured latency. Faster R-CNN V2 achieved nearly comparable average precision and a marginally higher F1 score, making it valuable as a secondary architecture for validation and thesis comparison. EfficientNetV2-M RetinaNet produced no usable evaluated detections and should be debugged before any further full-dataset training.

The most important class-level finding is that the system is much stronger on bodies and heads than on several tail classes. This has direct consequences for final fish-level grading: the downstream logic should preserve missing evidence rather than forcing a grade or rejection when a tail is not reliably detected. The next milestone is therefore not another immediate retraining cycle. It is controlled direct inference, new-image validation, full-frame-versus-crop integration testing, fish-part association evaluation, and finally an untouched end-to-end test on independent fish.

> **Current status in one sentence**
>
> Model 2 is no longer a "dead detector" at the development-evaluation level: YOLO26m and Faster R-CNN both learned the 12-class task. The remaining question is how well those detections generalize and survive the complete two-model application pipeline.

Appendix A. YOLO26m Exact Per-Class Development Metrics

Class	AP50	AP50-95
Grade_A_Body	0.994	0.655
Grade_A_Head	0.969	0.718
Grade_A_Tail	0.848	0.452
Grade_B_Body	0.737	0.336
Grade_B_Head	0.867	0.377
Grade_B_Tail	0.359	0.161
Grade_C_Body	0.970	0.651
Grade_C_Head	0.962	0.692

Class	AP50	AP50-95
Grade_C_Tail	0.646	0.270
Rejected_Body	0.865	0.490
Rejected_Head	0.820	0.536
Rejected_Tail	0.503	0.250

These values are development metrics from the completed YOLO26m run and should be reported as detector AP values, not final fish-level accuracy.

Appendix B. Quantitative Comparison Deltas

Comparison	Calculated difference	Interpretation
YOLO vs Faster mAP50-95	+2.02 percentage points for YOLO	Small but consistent YOLO advantage in averaged localization AP
YOLO vs Faster mAP50	+2.12 percentage points for YOLO	YOLO advantage at IoU 0.50
YOLO vs Faster mAP75	+0.71 percentage points for YOLO	Very small difference at stricter IoU
Precision	+1.04 percentage points for Faster R-CNN	Faster R-CNN slightly fewer false positives at its tuned threshold
Recall	+0.73 percentage points for YOLO	YOLO slightly fewer missed parts at its tuned threshold
F1	+0.21 percentage points for Faster R-CNN	Practically very close
Latency	2.26x slower for Faster R-CNN	Large practical difference for live processing